

Text Classification

Applied Text Mining, from Foundations to
Advanced

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From before

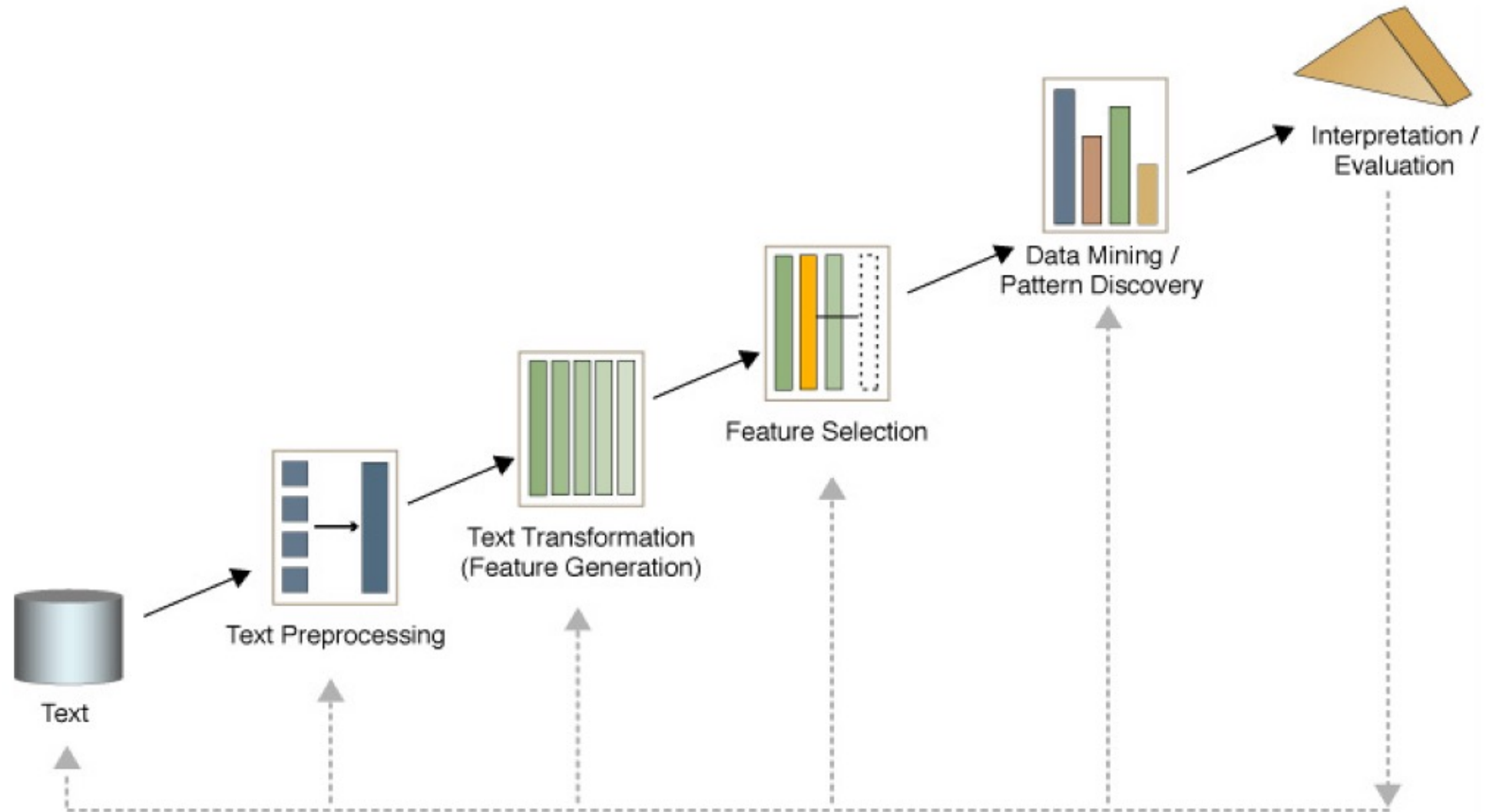
- Text preprocessing
- Tokenization
- Token removal
- Stemming
- Text representation

This lecture

- Text classification
- Classification pipeline
- Classification algorithms
- Evaluation

Text Classification

Text mining process



Text transformation, feature generation

Bag-of-Words representation

Doc1: Text mining is to identify useful information.

Doc2: Useful information is mined from text.

Doc3: Apple is delicious.

Document-Term matrix (DTM):

	text	information	identify	mining	mined	is	useful	to	from	apple	delicious
Doc1	1	1	1	1	0	1	1	1	0	0	0
Doc2	1	1	0	0	1	1	1	0	1	0	0
Doc3	0	0	0	0	0	1	0	0	0	1	1

Text classification

- **Supervised learning:** Learning a function that maps an input to an output based on example input-output pairs.
 - infer a function from labeled training data
 - use the inferred function to label new instances
- Human experts annotate a set of text data
 - Training set

Document	Class
Email1.....	Not spam
Email2....	Not spam
Email3....	Spam
...	...

Applications

- Assigning subject categories, topics, or genres
- Spam detection
- Authorship identification
- Age/gender identification
- Language Identification
- Sentiment analysis

Quiz?

- Which one **is not** a text classification task? (less likely to be)
 - Author's gender detection from text
 - Finding about the smoking conditions of patients from clinical letters
 - Grouping similar news articles
 - Classifying reviews into positive and negative sentiment

Terminology

- **Features:** information which is used to separate data into classes
- **Prediction:** the output of the model, which can be compared to the correct labels (our ground truth)
- **Parameters:** the data in the model
- **Training:** fitting parameters to the data
- **Hyperparameters:** the “settings” of the model; these are choices made by you

Types of classification

1. Binary classification:

- **Definition:** Involves two categories or classes.
- **Examples:** True/False, Yes/No, Spam/Not Spam, Positive/Negative sentiment.

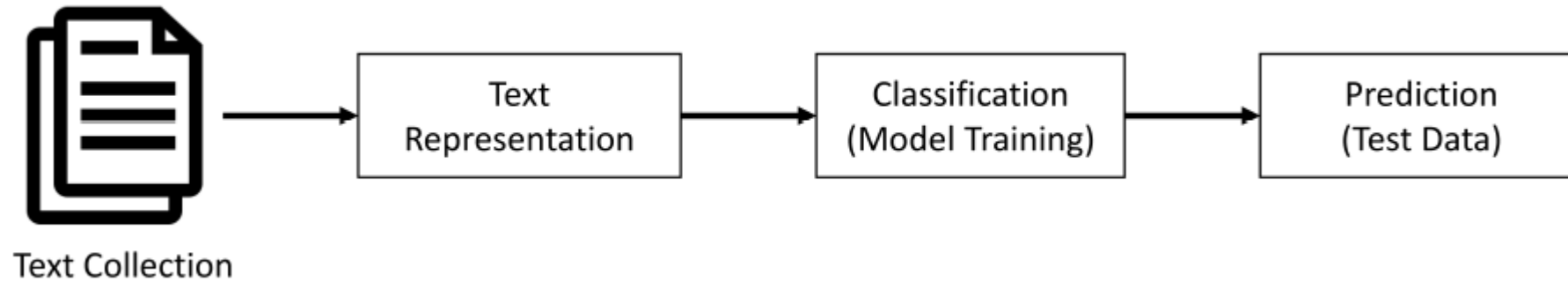
2. Multiclass classification:

- **Definition:** Involves more than two categories or classes.
- **Examples:** Classifying animals into Cats/Dogs/Birds, predicting the genre of a book (e.g., Fiction/Non-fiction/Science/History).

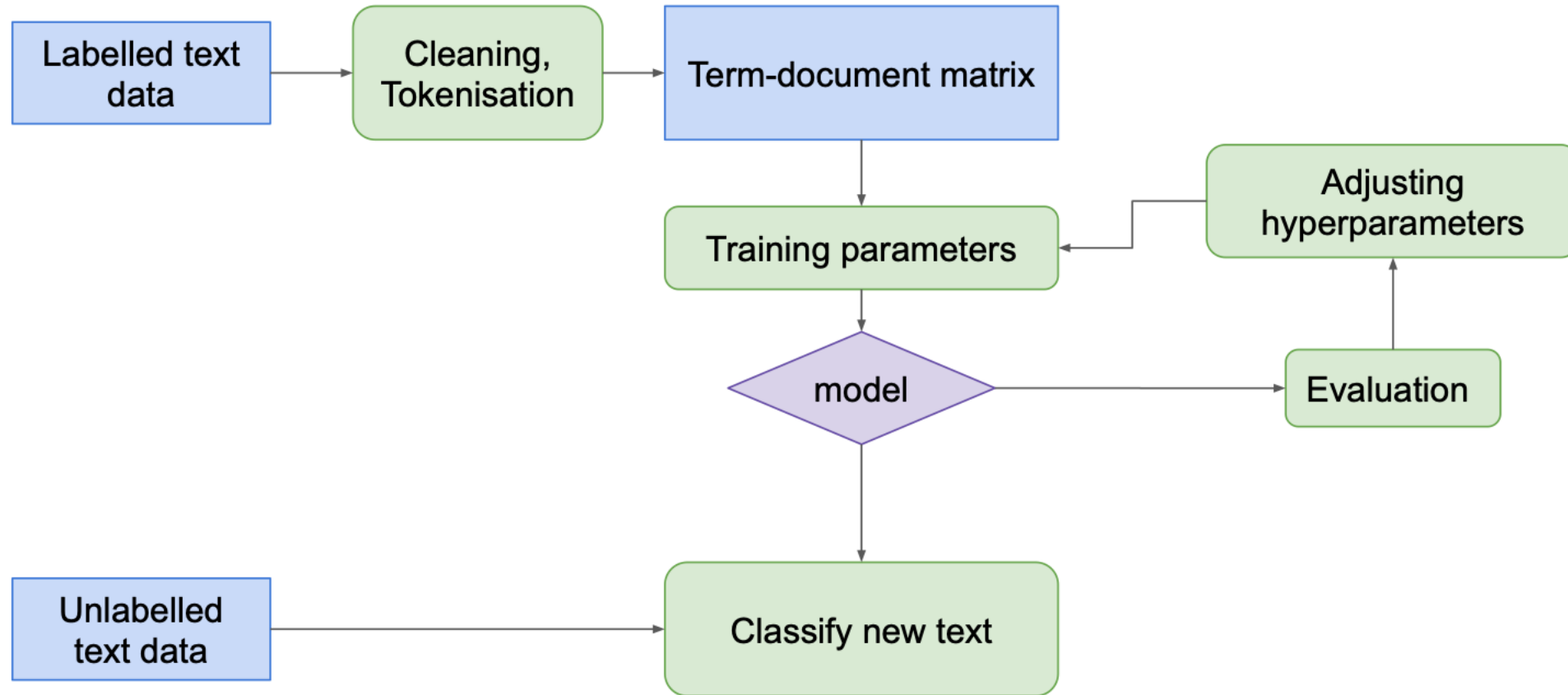
3. Multilabel classification:

- **Definition:** Each instance can belong to multiple classes simultaneously.
- **Examples:** Tagging a research article with multiple topics like "Machine Learning," "Natural Language Processing," and "Healthcare.", Assigning multiple genres to a book.

Simple pipeline



Opening the pipeline



Preparing data

- Data collection
- Text pre-processing
- Data splitting
 - Training
 - Validation (development)
 - to tune the hyperparameters
 - Test
- Text representation

Data splitting

- **Methods:**

- Train-Test Split

- Split the dataset into two parts: training and testing (e.g., 80%-20%).

- K-fold cross validation

- Split the dataset into K non-overlapping folds of approximately equal size.

- Leave-One-Out Cross-Validation

- A special case of K-Fold where $K=N$ (number of data points).

K-fold cross validation

1. Purpose:

- Evaluates the performance of a model on unseen data by splitting the dataset into K subsets (folds).
- Ensures that every data point gets a chance to be in both the training and validation sets.

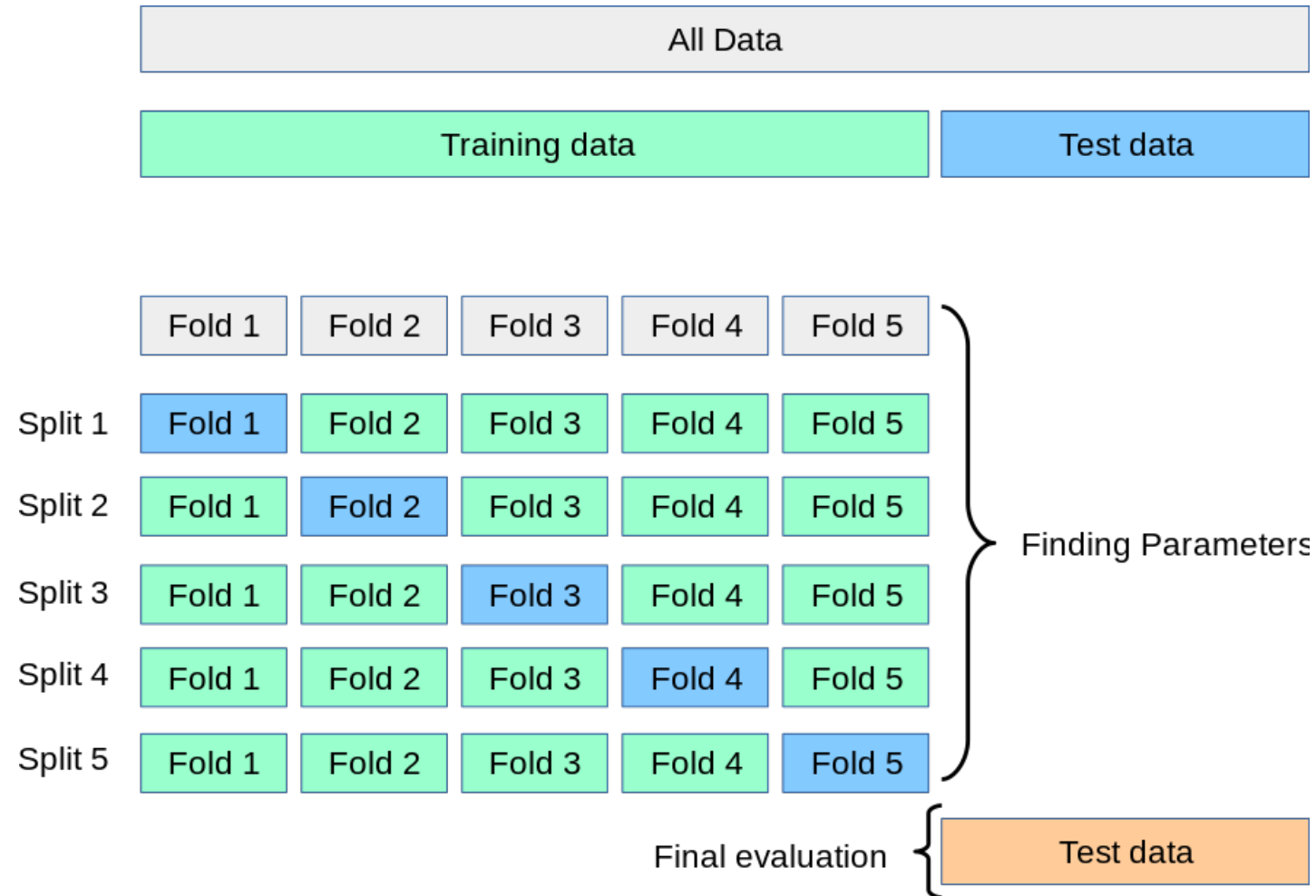
2. How it Works:

- Split the data into K equally-sized folds.
- Train the model on K-1 folds and validate it on the remaining fold.
- Repeat the process K times, each time using a different fold for validation.

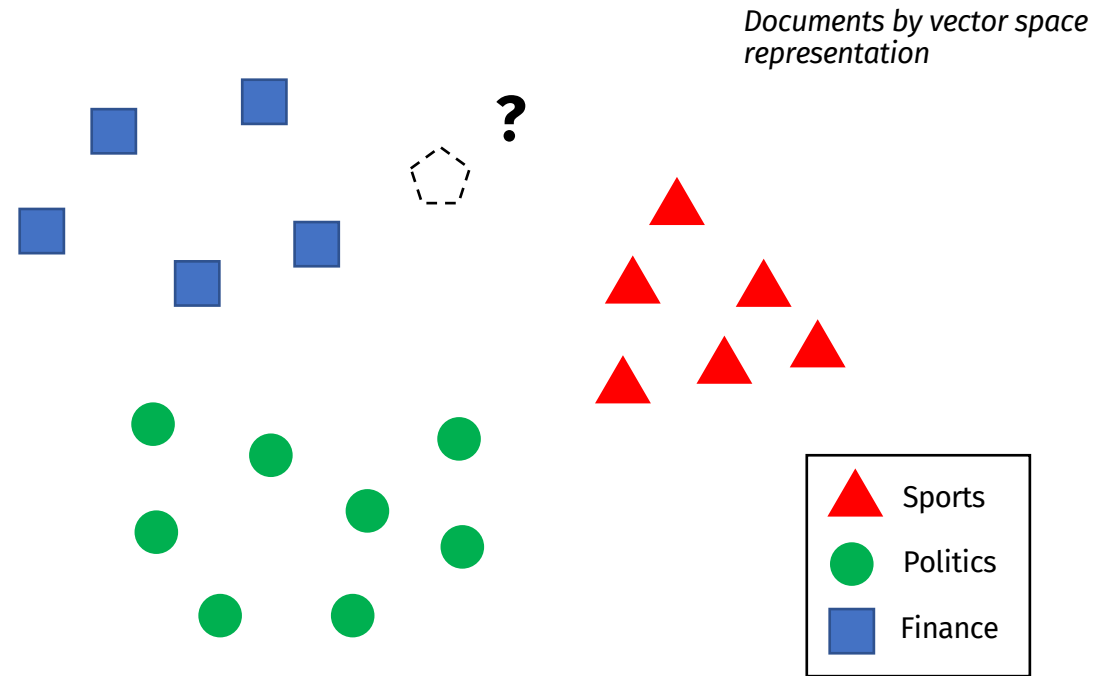
3. Advantages:

- Reduces the risk of overfitting compared to a single train-test split.
- Provides a more robust estimate of the model's generalization performance.

K-fold cross validation



How to classify this document?



Text Classification: definition

- *Input*:
 - a document d
 - a fixed set of classes $C = \{c_1, c_2, \dots, c_J\}$
- *Output*: a predicted class $c \in C$

Classification Algorithms

Hand-coded rules

- Rules based on combinations of words or other features
- Accuracy can be high: If rules carefully refined by expert
- But building and maintaining these rules is expensive
- Data/Domain specifics

Supervised Machine Learning

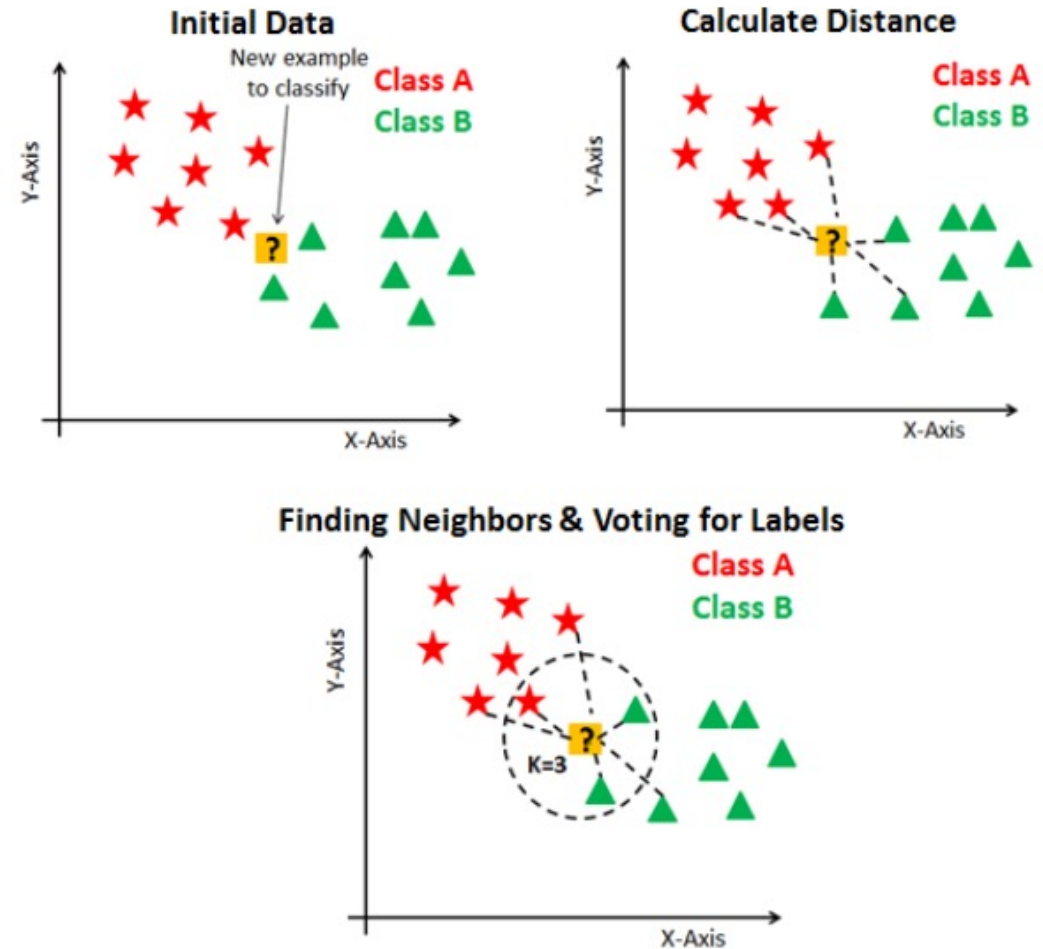
- *Input:*
 - a document d
 - a fixed set of classes $C = \{c_1, c_2, \dots, c_J\}$
 - A training set of m hand-labeled documents $(d_1, c_1), \dots, (d_m, c_m)$
- *Output:*
 - a learned classifier $y: d \rightarrow c$

Common algorithms

- K-nearest neighbors
- Naïve Bayes
- Logistic regression
- Support vector machines
- Neural networks
- Deep learning

K-nearest neighbors

- **K-Nearest Neighbors (KNN)** is a simple, non-parametric algorithm that classifies data points based on the majority class of their K nearest neighbors in feature space, using a distance metric such as Euclidean distance.
- Non-parametric methods: make no strong assumptions about the data's underlying distribution.



Naive Bayes

- **Naive Bayes** is a probabilistic machine learning algorithm based on Bayes' Theorem, which assumes that all features are independent of each other (a "naive" assumption) and uses this simplification to classify data efficiently.

$y(\$

great	2
love	2
recommend	1
laugh	1
happy	1
...	...

$\mathbf{)} = \mathbf{c}$

Bayes' rule

- For a document d and a class c

$$P(c | d) = \frac{P(d | c)P(c)}{P(d)}$$

Learning naive Bayes

$$c_{MAP} = \operatorname{argmax}_{c \in C} P(c \mid d)$$

MAP is “maximum a posteriori” = most likely class

$$= \operatorname{argmax}_{c \in C} \frac{P(d \mid c)P(c)}{P(d)}$$

Bayes Rule

$$= \operatorname{argmax}_{c \in C} P(d \mid c)P(c)$$

Dropping the denominator

Learning naive Bayes

$$c_{MAP} = \operatorname{argmax}_{c \in C} P(d | c)P(c)$$

Document d
represented as
features $x_1..x_n$

$$= \operatorname{argmax}_{c \in C} P(x_1, x_2, \dots, x_n | c)P(c)$$

Could only be estimated if a very, very large number of training examples was available.

We can just count the relative frequencies in a corpus

Learning naive Bayes

- Simply use the frequencies in the data

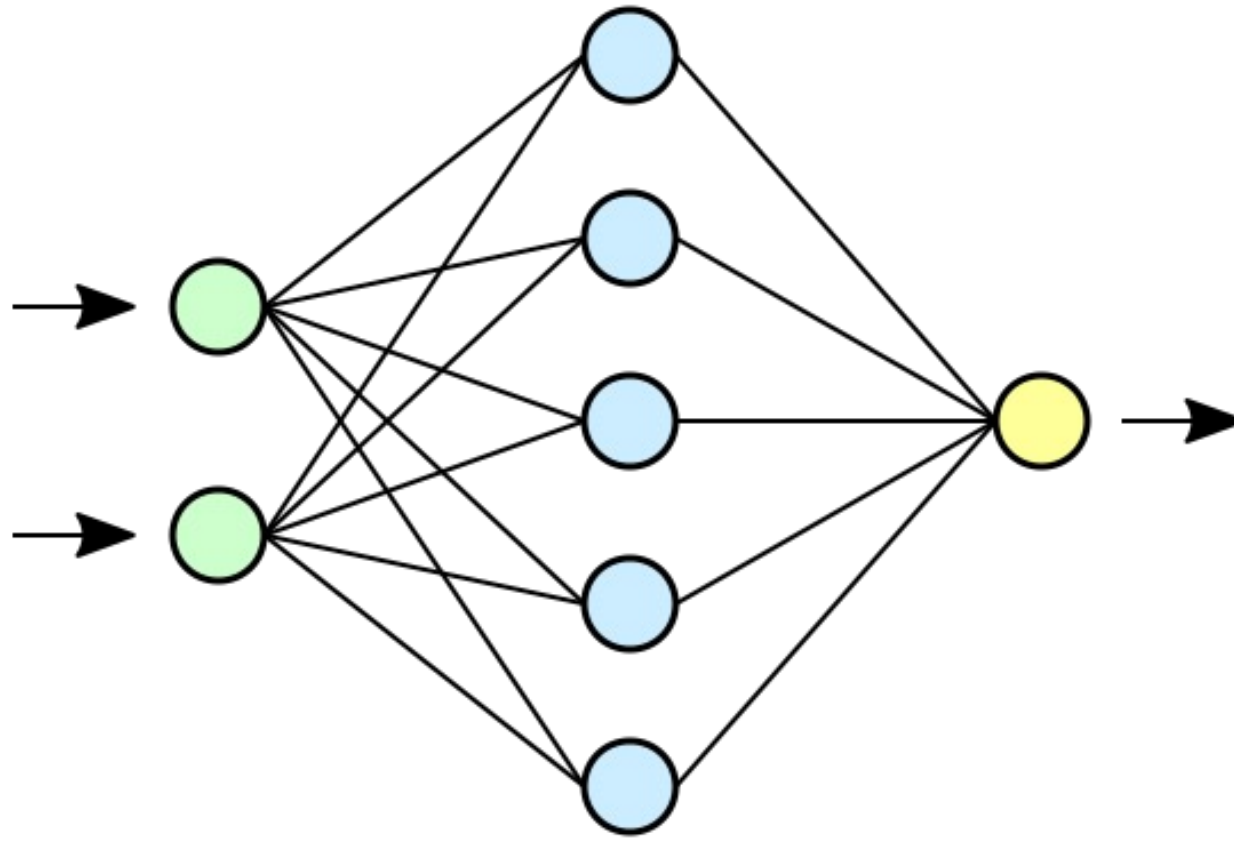
$$\hat{P}(c_j) = \frac{\text{doccount}(C = c_j)}{N_{doc}}$$

$$\hat{P}(w_i | c_j) = \frac{\text{count}(w_i, c_j)}{\sum_{w \in V} \text{count}(w, c_j)}$$

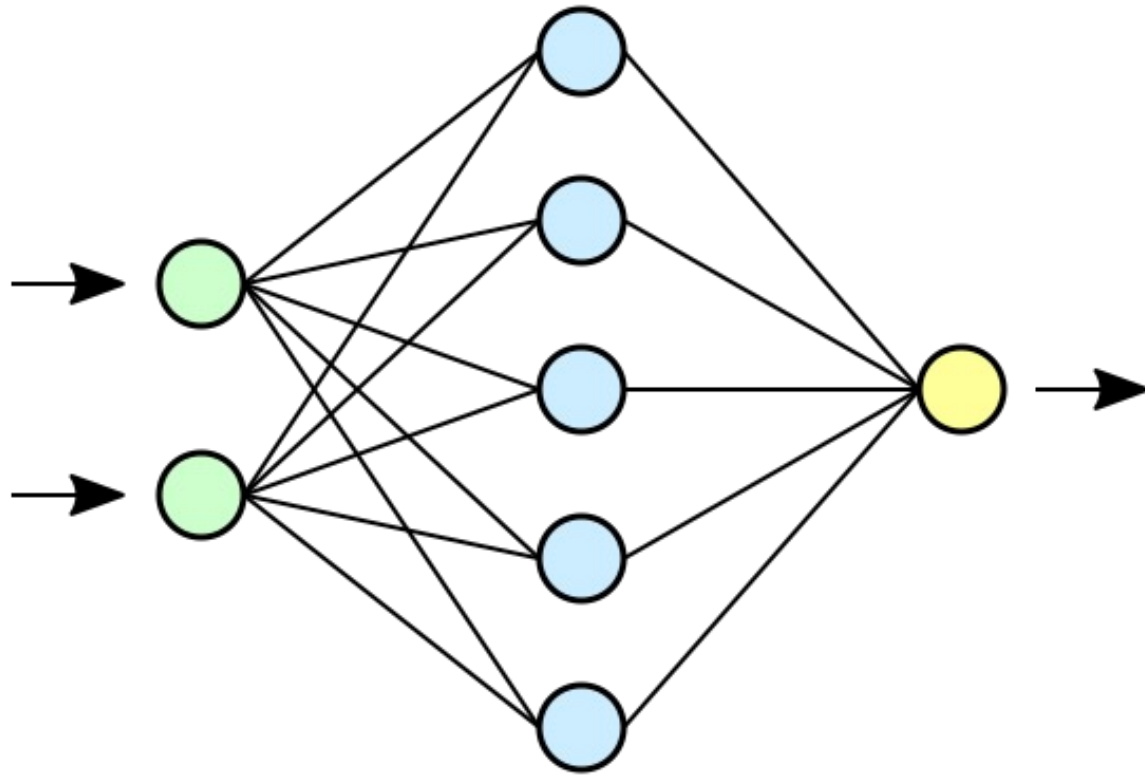
Neural networks, “deep learning”

- **Neural networks** are machine learning models inspired by the structure of the human brain, consisting of layers of interconnected nodes (neurons) that process and learn patterns in data by adjusting weights through a process called backpropagation.
 - Compositional approach to curve-fitting;
 - “Biologically inspired”
(but don’t take that too seriously);
 - Sound cool.

Neural network



Neural network



“Hidden” nodes:

Example:

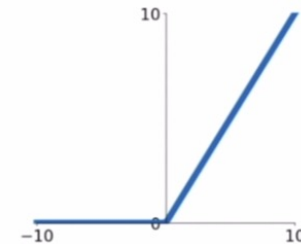
$$h_1 = f(w_{11}x_1 + w_{12}x_2)$$

Output:

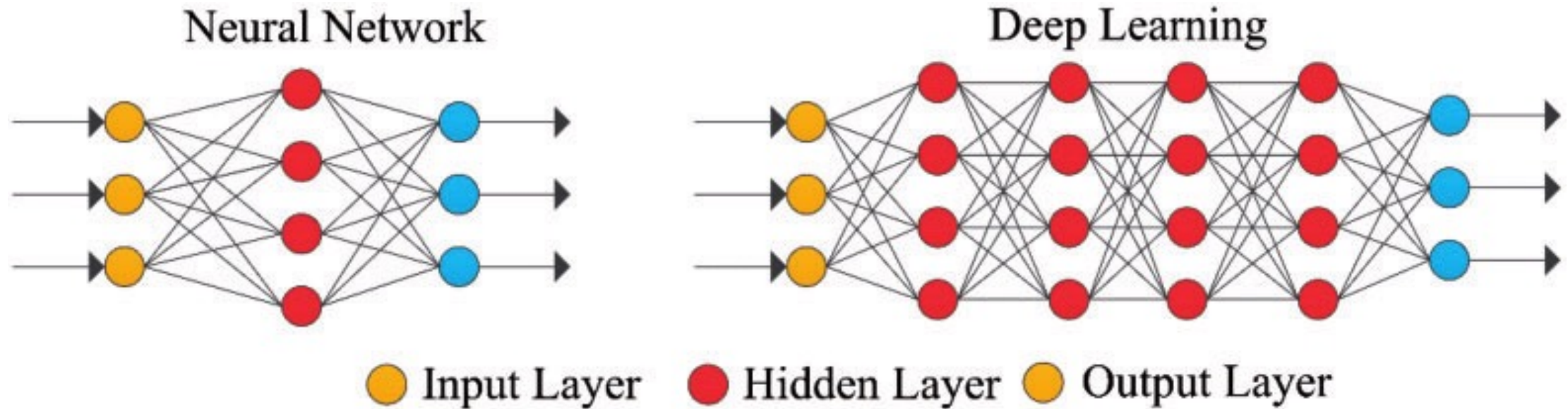
$$y = f(w_{21}h_1 + w_{22}h_2 + \dots + w_{25}h_5)$$

“Activation function”:

- ReLU $f(z) =$
- ...



What makes a neural net “deep”?



More on neural networks

- **Wednesday, Thursday and Friday!**

Ensemble classifiers

- We can combine multiple classifiers
 - **Random forest classifier**
 - fit multiple decision trees, average over predictions
 - Less “volatile” than a single decision tree
 - **Voting classifier**
 - Fit multiple classifiers and let them “vote” on the result
 - Classifiers may be of different classes!

Selecting hyperparameters

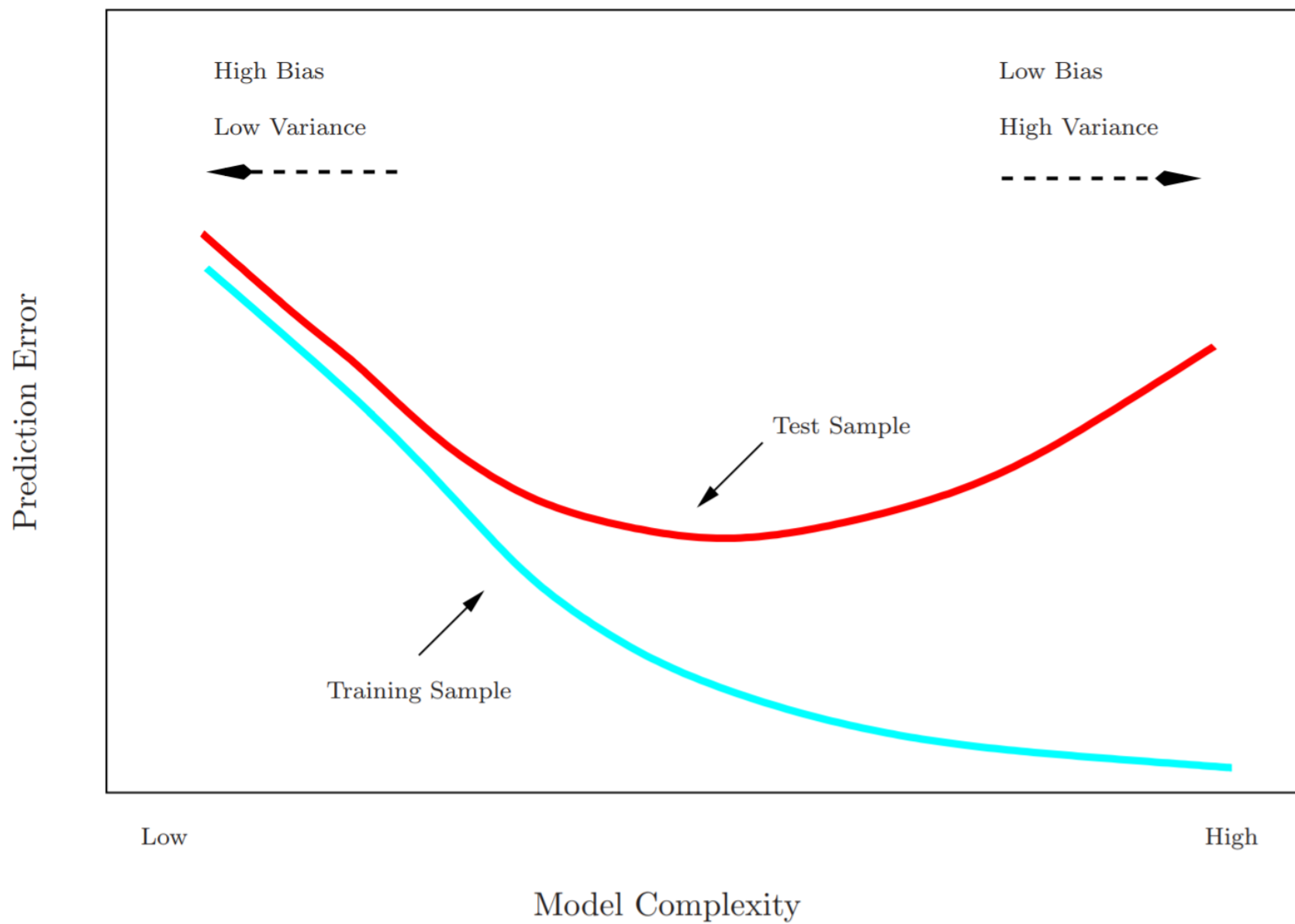
- A hyperparameter is a parameter whose value is used to control the learning process, which must be configured before the process starts.
 - Experts input
 - Hyperparameter optimization: the selection process will be automated
 - Random search
 - Grid search: systematically go through your data
 - Other optimization techniques

Evaluation

No free lunch

“Any two optimization algorithms are equivalent when their performance is averaged across all possible problems”

(Wolpert & MacReady)



Confusion matrix

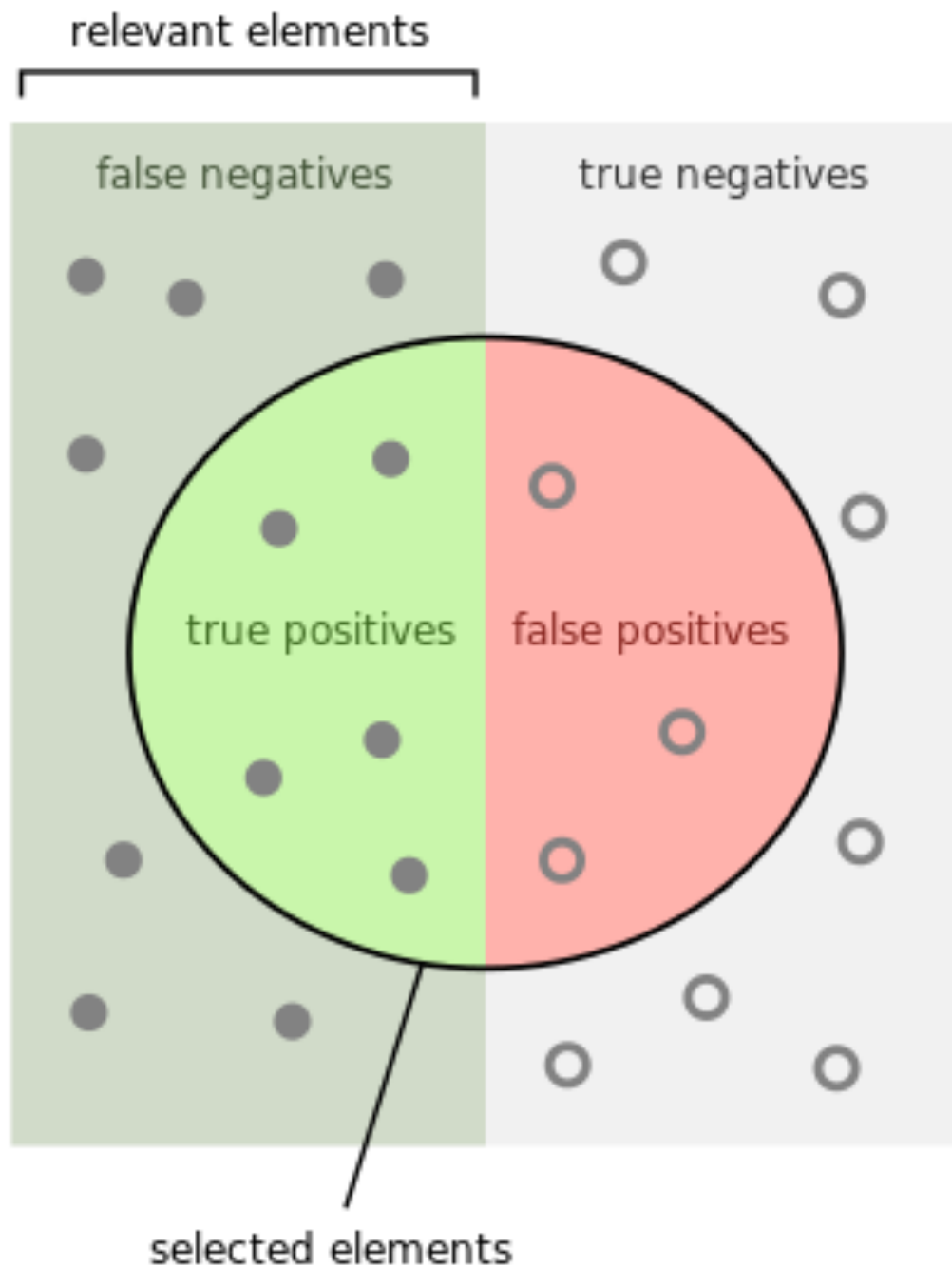
		Predicted Class		
		Positive	Negative	
Actual Class	Positive	True Positive (TP)	False Negative (FN) Type II Error	Sensitivity $\frac{TP}{(TP + FN)}$
	Negative	False Positive (FP) Type I Error	True Negative (TN)	Specificity $\frac{TN}{(TN + FP)}$
		Precision $\frac{TP}{(TP + FP)}$	Negative Predictive Value $\frac{TN}{(TN + FN)}$	Accuracy $\frac{TP + TN}{(TP + TN + FP + FN)}$

Accuracy

- Accuracy is a valid choice of evaluation for classification problems which are well balanced and not skewed.

Precision and recall

- **Precision:** % of selected items that are correct
Recall: % of correct items that are selected
- Precision is a valid choice of evaluation metric when we want to be very sure of our prediction.
- Recall is a valid choice of evaluation metric when we want to capture as many positives as possible.



How many selected items are relevant?

$$\text{Precision} = \frac{\text{true positives}}{\text{true positives} + \text{false positives}}$$

How many relevant items are selected?

$$\text{Recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$$

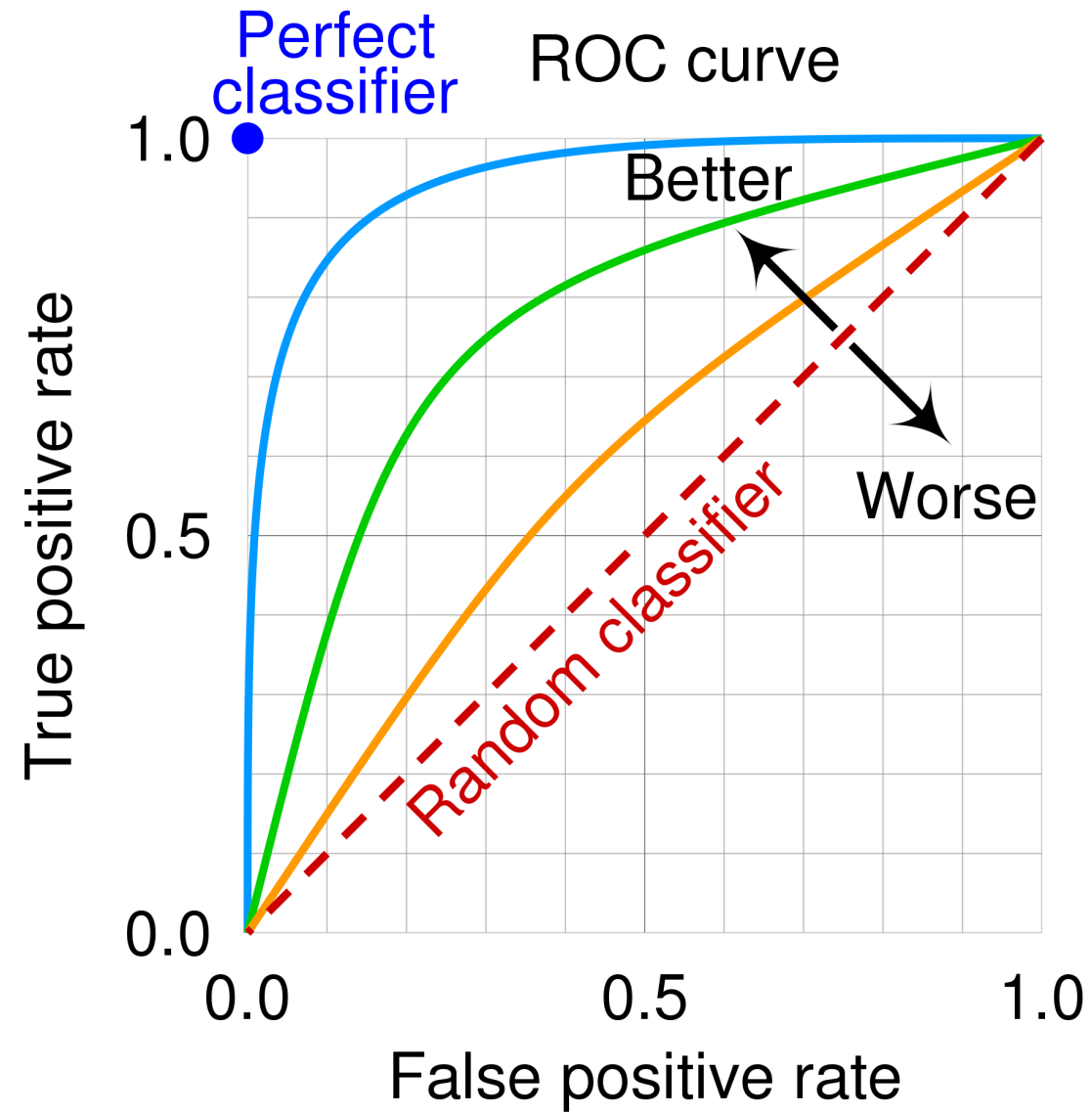
Source: <https://en.wikipedia.org/wiki/F-score>

A combined measure: F

- A combined measure that assesses the P/R tradeoff is F measure (weighted harmonic mean):

$$F = \frac{1}{\alpha \frac{1}{P} + (1 - \alpha) \frac{1}{R}} = \frac{(\beta^2 + 1)PR}{\beta^2 P + R}$$

- The harmonic mean is a very conservative average
- People usually use balanced F1 measure
 - i.e., with $\beta = 1$ (that is, $\alpha = 1/2$): $F = 2PR/(P+R)$



Evaluation metrics: multiclass classification

- Evaluation is less straightforward with multiple classes
- Precision/recall/F1 per class
- But what if we want a single score?
 - **Macro F1**: average of F1 score for each class
 - **Micro F1**: uses total number of true/false positives/negatives

Conclusion

- ❖ Understanding the classification pipeline is essential for building effective models, from preprocessing text data to training and evaluation.
- ❖ There are different algorithms available for text classification, each with its strengths.
- ❖ Accurate evaluation using metrics like accuracy, precision, recall, and cross-validation ensures reliable model performance and generalization to unseen data.

Practical 2

Questions?